# Curriculum Vitae


## PUNEET PAREEK

Tata Institute of Fundamental Research Hyderabad, India
Email : ppareek@tifrh.res.in


## Research Interests

Glassy Dynamics

Disordered Systems

Active Matter

Non-equilibrium Statistical Mechanics

Biophysics

## Education

**Ph.D.** 2020-2026

University: Tata Institute of Fundamental Research (TIFR), Mumbai, India

Thesis Title: *Characteristics of glassy dynamics and how active self-propelled forces affect them*

Supervisor: Prof. Saroj Kumar Nandi, TIFR Hyderabad, India

**M.Sc. Physics** 2018-2020

University: Indian Institute of Technology Delhi (IIT-D), New Delhi, India

Project Title: *Shortcuts to adiabaticity with active particles*

Supervisor: Prof. Rahul Suresh Marathe, IIT-D, India

CGPA: 9.507/10.000

**B.Sc. (H) Physics** 2015-2018

University: University of Delhi, Delhi, India

CGPA: 9.203/10.000

# Professional Experience

**Research Associate** Oct 2025 - Apr 2026

University: Tata Institute of Fundamental Research (TIFR), Mumbai, India

Supervisor: Prof. Saroj Kumar Nandi, TIFR Hyderabad, India

**Postdoc** Apr 2026 - Sep 2026

University: Tata Institute of Fundamental Research (TIFR), Mumbai, India

Supervisor: Prof. Saroj Kumar Nandi, TIFR Hyderabad, India

# Statement of Past Research

- **Various time scales in passive glassy dynamics**: We found that caging controls the mean, whereas dynamical heterogeneity (DH) governs the distribution of particle displacements. In addition, we can define several other time scales, such as when the mean-square displacement becomes diffusive, the distribution of particle displacement becomes Gaussian, the overlap function becomes a power-law, the intermediate scattering function becomes exponential, etc. We demonstrated using analytical calculations and simulation results that all these different timescales are related to either caging or DH. This naturally explains the Fickian yet non-Gaussian behaviour of glasses.

- **Crossover in autocorrelation function from stretched exponential to power-law**: The overlap function in passive glasses shows a crossover from stretched exponential (SE) at short times to a power law at long times. To understand the mechanism behind such a crossover, we defined a simplified one-dimensional spin model that belongs to the class of kinetically constrained models. The model has an onsite energy function and a kinetic rule that a spin can flip only if the neighbouring spins have opposite signs. Thus, the dynamics conserve energy, which comes from the movements of domain walls (DWs). We found that the crossover comes from a change in the relevant excitations: from DWs at short times to bound pairs of DWs at long times.

- **Influence of activity on relaxation dynamics and fragility in active glasses**: Active glasses comprise self-propelled particles with a self-propulsion force $f_0$ and persistence time $\tau_p$ and provide a minimal model for biological systems. They often show nontrivial behaviors, such as reentrant dynamics and tunable fragility. We studied a dense active system of soft repulsive particles that shows these features. We characterised the critical surface separating liquid and glassy states and showed that the nontrivial features are consequences of the morphology of this surface. In addition, with increasing $\tau_p$, the nature of the critical point changes from glass-like to jamming-like. The tunable fragility, where the dynamics goes from sub-Arrhenius to super-Arrhenius, also appears in biological tissues. We employed a dynamic scaling analysis and showed that this behavior in the current model is governed by an "active glass line".

## Conferences Attended

- In-house Symposium, 15 – 17 October, 2022

  Tata Institute of Fundamental Research Hyderabad, India

  Poster title: *Stretched Exponential to Power Law in Glassy Systems.*

- Frontiers in Active and Soft Matter 2023, 10 and 11 February, 2023

  Tata Institute of Fundamental Research Hyderabad, India

  Poster title: *Can Glasses be Fickian yet non-Gaussian?*

- Physics of Life: Active and Living Matter, 8-10 February, 2024

  Institute of Physics, Bhubaneswar, India

  Poster title: *Re-entrance of Relaxation Dynamics in Dense Assemblies of Active Particles.*

- The Discussion Meeting on Glassy, Disordered and Non-equilibrium Soft Matter, 10-11 April, 2025

  Raman Research Institute, Bangalore, India

  Poster title: *Scaling the glassy dynamics of active particles: Tunable fragility and reentrance.*

## Awards and Scholarships

- 2017-18: *Science Meritious Award* at University of Delhi.
- 2019-20: *Merit-Cum-Mean* and *Prof. Vidya Bhushan Scholarship* at Indian Institute of Technology, Delhi.
- 2023: *Best Poster Award* in the conference ‘‘Frontiers in Active and Soft Matter’’ (2023).

## Achievements

- 2020: Graduate Aptitude Test in Engineering (GATE) Physics with All India Rank-7
- 2020: Joint Entrance Screening Test (JEST) with All India Rank-59
- 2020: Junior Research Fellowship (JRF) Physics with All India Rank-76
- 2018: Joint Admission Test (JAM) Physics with All India Rank-106

## Software and Computational Skills

**Programming skills:** Python, C, C++

**Software:** Mathematica, LATEX, GNU plot, LAMMPS, OVITO

**Operating systems:** Linux, Microsoft Windows

# List of Publications

**Published**

- **Puneet Pareek**, Monoj Adhikari, Chandan Dasgupta, Saroj Kumar Nandi, *Different glassy characteristics are related to either caging or dynamical heterogeneity*, J. Chem. Phys. 159, 174503 (2023).

- Sukanta Mukherjee, **Puneet Pareek**, Mustansir Barma, Saroj Kumar Nandi, *Stretched exponential to power-law: crossover of relaxation in a kinetically constrained model*, J. Stat. Mech. (2024) 023205.

- **Puneet Pareek**, Peter Sollich, Saroj Kumar Nandi, Ludovic Berthier, *Scaling the glassy dynamics of active particles: Tunable fragility and reentrance*, Proc. Natl. Acad. Sci. U.S.A. 123 (4) e2516624123 (2025).

**Under Review**

- Soumitra Kolya, **Puneet Pareek**, Saroj Kumar Nandi, *Active Inhomogeneous Mode-Coupling Theory (aIMCT) for Dense Systems of Self-Propelled Particles*, arXiv:2410.15928, (2024).

**Under Preparation**

- **Puneet Pareek**, Saroj Kumar Nandi, *Non trivial effect of fraction of active particles on glassy dynamics.*

# Referees

1. **Dr. Saroj Kumar Nandi**

   Reader-F

   Tata Institute of Fundamental Research Hyderabad, India

   Email: saroj@tifrh.res.in

2. **Prof. Chandan Dasgupta**

   Honorary Professor

   Indian Institute of Science, Bangalore, India

   ICTS Endowed Visiting Professor

   International Centre for Theoretical Sciences, Bangalore, India

   Email: cdgupta@iisc.ac.in

3. **Prof. Mustansir Barma**

   Professor Emeritus

   Tata Institute of Fundamental Research Hyderabad, India

   Email: barma@tifrh.res.in